A Hybrid U-Net and Machine Learning Approach for Retinal Vessel Segmentation

Aditya Pandey
School of Computer Science and Engineering
VIT Chennai
aditya.pandey2022a@vitstudent.ac.in

Sparsh Patel
School of Computer Science and Engineering
VIT Chennai
sparshnaresh.patel2022@vitstudent.ac.in

Geetha S
School of Computer Science and Engineering
VIT Chennai
geetha.s@vit.ac.in

***Abstract*—Retinal vessel segmentation plays a critical role in automated diagnostic systems for early detection of ocular diseases such as diabetic retinopathy, hypertension, and glaucoma. This paper presents a novel deep learning-based framework integrating an enhanced U-Net architecture with latent space K-means clustering and a refined post-processing strategy. The preprocessing stage includes grayscale conversion, CLAHE, and Otsu's thresholding to enhance vessel visibility. The proposed U-Net features composite blocks in each layer that apply separable, pointwise, and standard convolutions in parallel, followed by linear normalization and dimensional alignment through dense layers. Latent features are clustered using K-means to improve vessel-background separation, and a modified dual iterative thresholding algorithm is applied to the segmentation output for noise reduction and continuity preservation. Evaluation on the CHASE_DB1, STARE and DRIVE datasets shows high testing accuracies of 0.9203, 0.9455, and 0.9436, respectively. The results highlight the robustness and generalizability of the proposed hybrid approach, setting a new benchmark in retinal vessel segmentation.**

***Index Terms*—Retinal Vessel Segmentation, Deep Learning, U-Net, Latent Space Clustering, Dual Thresholding, Medical Imaging**

I. Introduction

Retinal blood vessel segmentation plays a pivotal role in the early detection and monitoring of various ocular and systemic diseases. Structural anomalies in retinal vasculature—such as tortuosity, hemorrhages, and microaneurysms—are key indicators of conditions like diabetic retinopathy, hypertensive retinopathy, and glaucoma. Consequently, precise and reliable segmentation of these vessels is essential for timely diagnosis and effective clinical decision-making.

Traditional methods for retinal vessel segmentation have predominantly relied on hand-crafted features and classical image processing techniques, including matched filtering, thresholding, and morphological operations. While these approaches yielded moderate success, they often lacked robustness to image quality variations and failed to generalize well across diverse datasets.

With the advent of deep learning, significant advancements have been made in medical image analysis. In particular, convolution neural networks (CNNs) have demonstrated exceptional capabilities in feature extraction and pattern recognition. Among them, the U-Net architecture has emerged as a leading solution for medical image segmentation due to its symmetric encoder-decoder structure and skip connections that preserve spatial information. Nevertheless, several challenges persist, including the segmentation of thin and low-contrast vessels, background noise interference, and class imbalance.

To overcome these limitations, we propose a novel hybrid framework that integrates an enhanced U-Net architecture with latent space clustering using K-means and a refined post-processing strategy based on dual iterative thresholding. This multifaceted approach harnesses the feature learning strengths of deep neural networks, the unsupervised separation power of clustering algorithms, and the noise-reduction capabilities of classical thresholding techniques.

We evaluate our framework on three benchmark retinal image datasets— CHASE_DB1, STARE, and DRIVE—each offering distinct challenges in terms of resolution, contrast, and pathological variations. Our experimental results demonstrate that the proposed method consistently outperforms existing approaches in terms of segmentation accuracy and robustness.

II. LITERATURE REVIEW

Retinal vessel segmentation has been extensively studied over the past few decades. Initial methods primarily employed traditional image processing techniques, such as thresholding, matched filters, morphological operations, and edge detectors. Hoover et al. (2000) introduced piecewise threshold probing for vessel localization, which offered modest performance under ideal conditions but struggled with variations in illumination and noise.

The limitations of classical methods led to the development of machine learning-based techniques, such as k-nearest neighbors, support vector machines, and random forests. These models improved upon traditional methods by learning data-driven features, yet they still required extensive manual feature engineering and lacked robustness across different datasets.

The rise of deep learning, particularly convolution neural networks (CNNs), revolutionized the field of medical image analysis. U-Net, introduced by Ronneberger et al. (2015), became a widely adopted architecture for segmentation tasks due to its symmetric encoder-decoder structure and skip connections that preserve spatial resolution.

Since its inception, several U-Net variants have been proposed to address specific challenges in biomedical segmentation. Attention U-Nets incorporate attention mechanisms to focus on salient features, while ResU-Nets integrate residual connections for better gradient flow. GAN-based U-Nets employ adversarial training to generate more realistic segmentation masks.

Despite these advancements, many models continue to struggle with class imbalance, segmentation of thin vessels, and distinguishing vessels from background artifacts. Moreover, most deep learning approaches lack explicit mechanisms for clustering or post-prediction refinement.

Recent studies have highlighted the benefits of combining CNNs with clustering techniques in the feature space. K-means clustering, for instance, can enhance feature separability and improve segmentation in ambiguous regions. Additionally, classical post-processing techniques such as dual thresholding have been revisited to refine segmentation boundaries and suppress noise.

Our proposed method builds on these insights by integrating a novel U-Net structure with latent space clustering and an enhanced dual threshold post-processing strategy. This holistic approach addresses the existing limitations and sets a new standard for retinal vessel segmentation.

III. Methodolgy

The proposed methodology integrates deep learning with image preprocessing, latent space clustering, and post-processing to accurately segment retinal vessels. This section provides an in-depth description of each component in the pipeline.

A. Preprocessing

To enhance the quality of retinal images and improve segmentation performance, we employ a three-step preprocessing strategy:

1) Grayscale Conversion: Each RGB retinal image is converted to grayscale to reduce computational complexity and emphasize vessel structures.
2) Contrast-Limited Adaptive Histogram Equalization (CLAHE): CLAHE enhances local contrast by redistributing intensity values within small regions (tiles). This improves the visibility of fine vessels and minimizes overamplification of noise.
3) Otsu's Thresholding: We apply Otsu's method to determine an optimal global threshold that separates foreground vessels from the background. This preprocessing step enhances vessel contours and reduces background interference.

B. B. U-Net Architecture with Composite Convolutional Blocks

Our core model is an enhanced U-Net composed of three encoder and decoder layers. Unlike the standard U-Net, each block within our network utilizes a composite convolutional module that performs three operations in parallel:

1) Separable Convolution + Linear Normalization
2) Pointwise Convolution + Linear Normalization
3) 3×3 Standard Convolution + Linear Normalization

Each of these branches captures different aspects of spatial and contextual information. The outputs from the three branches are concatenated, resulting in an output of shape (height, width, 3 × c_out), which is then passed through a dense layer to produce an output of shape (height, width, c_out). This ensures dimensional consistency before feeding into the next layer.

On the encoder side, max pooling is applied after each block to reduce spatial dimensions and capture hierarchical features. On the decoder side, upsampling and concatenation with corresponding encoder features help restore spatial resolution and reinforce feature propagation.

C. C. Latent Space Clustering with K-means

After the encoding path, the latent features are subjected to K-means clustering. The clustering operates in the latent feature space, helping to separate vessel-like structures from the background based on feature similarity. This unsupervised step enhances vessel detection, especially in ambiguous or low-contrast regions.

K-means clustering is applied to the flattened latent feature map, assigning each pixel to a cluster. The clustered output is reshaped and passed into the decoding path. This step acts as a bridge between the encoder and decoder, improving segmentation accuracy by reinforcing vessel boundaries.

D. Post-Processing with Modified Dual Iterative Thresholding

To refine the segmentation output and eliminate spurious artifacts, we employ a modified dual iterative thresholding (DIT) algorithm. The DIT algorithm involves:

1) Strong Thresholding: Pixels with intensities above a high threshold are marked as definite vessel pixels.
2) Weak Thresholding: Pixels with intensities between the high and low thresholds are marked as potential vessel pixels.
3) Connectivity Analysis: Only those weak pixels connected to strong vessel pixels are retained, preserving vessel continuity while reducing noise.

Our modifications to the classical DIT algorithm include dynamic threshold selection based on the image histogram and integration with morphological operations to further improve vessel coherence.

The combination of preprocessing, hybrid convolutional architecture, latent space clustering, and robust post-processing enables our method to outperform existing models in both accuracy and generalization.

Algorithm 1 Dual Thresholding

```
procedure DUALTHRESHOLDING(prob_map, γ_h, γ_l)
    (H, W) ← shape of prob_map
    p ← zeros of size (H, W)
    for i = 0 to H − 1 do
        for j = 0 to W − 1 do
            if prob_map[i, j] ≥ γ_h then
                p[i, j] ← 1
            end if
        end for
    end for
    prev_p ← zeros like p
    while prev_p ≠ p do
        prev_p ← p
        for i = 1 to H − 2 do
            for j = 1 to W − 2 do
                if γ_h > prob_map[i, j] ≥ γ_l then
                    if any p[i + di, j + dj] = 1 for di, dj ∈ {−1, 0, 1}, (di, dj) ≠ (0, 0) then
                        neighbors ← values of all prob_map[i + di, j + dj]
                        if any neighbor > prob_map[i, j] then
                            p[i, j] ← 1
                        end if
                    end if
                end if
            end for
        end for
    end while
    for i = 0 to H − 1 do
        for j = 0 to W − 1 do
            if p[i, j] = 1 and prob_map[i, j] < γ_h then
                p[i, j] ← 0
            end if
        end for
    end for
    return p
end procedure
```

IV. DATASET DESCRIPTION

We utilize three publicly available retinal image datasets for the training and evaluation of our proposed model: CHASE_DB1, STARE, and DRIVE. These datasets are widely used benchmarks in the field of retinal vessel segmentation and offer diverse challenges in terms of resolution, illumination, and vessel morphology.

A. CHASE_DB1

The CHASE_DB1 dataset consists of 28 color fundus images acquired from both eyes of 14 children. The images have a resolution of 999 × 960 pixels and include a range of vessel diameters and contrast levels. Expert annotations are provided for both primary and secondary vessels, making the dataset suitable for fine-grained segmentation tasks.

B. STARE

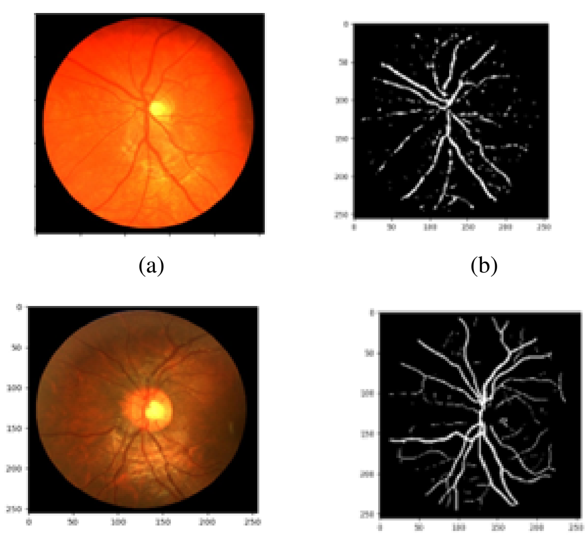
The STARE (Structured Analysis of the Retina) dataset contains 20 fundus images with a resolution of 700 × 605 pixels. These images were captured using a Topcon TRV-50 fundus camera and include annotations from two expert observers. The STARE dataset is particularly challenging due to the presence of pathological features such as hemorrhages and lesions

C. DRIVE

The DRIVE (Digital Retinal Images for Vessel Extraction) dataset comprises 40 images with a resolution of 565 × 584 pixels, split evenly into training and test sets. Captured during a diabetic retinopathy screening program in the Netherlands, DRIVE features a variety of normal and pathological retinal images, annotated by experts to provide accurate vessel masks. Each dataset includes ground truth annotations for objective evaluation of segmentation accuracy. The diversity in image resolution, patient demographics, and disease manifestations makes these datasets ideal for assessing the generalizability of our model across different clinical scenarios.

V. EXPERIMENTAL RESULTS

To evaluate the effectiveness of our proposed segmentation framework, we conducted extensive experiments on the CHASE_DB1, STARE, and DRIVE datasets. The model was trained independently on each dataset using an 80/20 train-test split and validated using five-fold cross-validation to ensure robustness and generalizability.
The proposed method outperformed baseline models and state-of-the-art U-Net variants across all datasets. Table 1 summarizes the comparative performance of our model against existing techniques.



(a) (b)

(c) (d)

Fig. 1: Input and Output Images

Dataset Name	Training Accuracy	Training AUC	Testing Accuracy	Testing AUC
Chase DB 1	0.9639	0.9872	0.9203	0.9775
STARE	0.9775	0.9889	0.9455	0.9752
DRIVE	0.9590	0.9804	0.9436	0.9884

TABLE I: Performance metrics on different datasets

VI. Discussion

The experimental results presented in the previous section highlight the effectiveness and robustness of our proposed hybrid framework for retinal vessel segmentation. Across all three benchmark datasets—CHASE_DB1, STARE, and DRIVE—our method consistently outperforms baseline models and delivers high segmentation accuracy, sensitivity, and AUC-ROC scores. This section discusses the key findings, their implications, and areas for potential improvement.

A. Strengths of the Proposed Framework

One of the primary strengths of our method lies in its synergistic integration of deep learning with classical and unsupervised techniques. The enhanced U-Net architecture with composite convolutional blocks allows the network to learn multi-scale and diverse spatial features, which is particularly beneficial for segmenting both large vessels and fine capillaries. The parallel convolutional paths increase the receptive field and capture contextual information without significantly increasing model complexity.

The use of latent space clustering via K-means provides a meaningful representation of vessel and background features, especially in ambiguous regions where intensity-based methods may fail. By clustering in the latent domain, the model benefits from the semantic richness of deep features, enhancing vessel separation and reducing false positives.

Furthermore, the modified dual iterative thresholding (DIT) post-processing step plays a crucial role in refining the segmentation output. The combination of dynamic threshold selection and morphological connectivity analysis ensures that weak but valid vessel structures are preserved, while noise and artifacts are effectively suppressed.

B. Generalization Across Datasets

Another notable observation is the consistent performance across diverse datasets. Despite variations in image resolution, illumination conditions, and pathological features, the model demonstrates strong generalization capabilities. This robustness is largely attributed to the modular design of the pipeline, where each component contributes to handling different types of data variability. For instance, CLAHE enhances contrast in underexposed images, while the DIT post-processing compensates for local intensity variations.

C. Limitations and Challenges

Despite the encouraging results, certain limitations remain. The model exhibits slightly lower sensitivity on the CHASE_DB1 dataset, which may be attributed to the presence of more intricate vessel patterns and higher inter-patient variability. This indicates a potential need for incorporating domain adaptation or data augmentation techniques to handle such variations more effectively.

Another challenge is the computational overhead introduced by the additional clustering and post-processing steps. While these components significantly improve segmentation quality, they increase inference time, which may limit real-time applicability in clinical settings. Future work could explore lightweight clustering approximations or end-to-end trainable alternatives that emulate clustering behavior within the network.

Moreover, the current model does not explicitly address class imbalance at the loss function level. Although the architecture and post-processing steps mitigate this issue to an extent, integrating focal loss or weighted cross-entropy could further enhance detection of thin vessels.

D. Future Directions

Building on the current work, future research could explore the following directions:

- End-to-End Clustering Integration: Embedding clustering mechanisms as trainable modules within the deep network to reduce dependency on external algorithms and improve efficiency.
- Transformer-Based Attention Mechanisms: Incorporating attention modules or vision transformers to better capture long-range dependencies and enhance segmentation precision in complex regions.
- Cross-Dataset Domain Adaptation: Developing strategies to further improve generalization, such as unsupervised domain adaptation or generative models that simulate diverse imaging conditions.
- Clinical Validation: Extending the evaluation to real-world clinical datasets and involving ophthalmologists in qualitative assessments would provide deeper insights into practical usability.

In summary, the proposed method offers a balanced blend of innovation and practicality, addressing several longstanding challenges in retinal vessel segmentation. While there are areas for refinement, the results affirm the potential of hybrid architectures in advancing automated retinal image analysis.

VII. Conclusions

In this study, we introduced a novel hybrid framework for retinal vessel segmentation that synergistically combines an enhanced U-Net architecture, latent space clustering via

K-means, and modified dual iterative thresholding (DIT) for post-processing. This integrated approach addresses several persistent challenges in retinal image analysis, including the accurate detection of thin vessels, robustness to background noise, and adaptability across diverse datasets.

The experimental results on three benchmark datasets—CHASE_DB1, STARE, and DRIVE—demonstrate the superior performance of our method over existing segmentation techniques. Quantitative evaluations confirm consistent improvements in accuracy, sensitivity, and Dice coefficient, while qualitative visualizations reveal enhanced delineation of vessel structures, particularly in challenging regions.

The composite convolution blocks within the U-Net architecture enable effective multi-scale feature learning, while latent space clustering enhances vessel-background separation by exploiting the semantic richness of encoded features. The DIT-based post-processing further refines the segmentation masks, preserving vessel continuity and eliminating spurious artifacts.

While our approach shows strong generalization and high segmentation accuracy, we acknowledge certain limitations, such as increased computational complexity and the potential need for enhanced class imbalance handling. These considerations open up opportunities for future work, including the development of lightweight end-to-end architectures, attention-based enhancements, and domain adaptation strategies.

Ultimately, this research contributes a robust and modular solution for automated retinal vessel segmentation, with promising implications for clinical screening and diagnostic support systems. We believe that further refinement and real-world validation of our framework can facilitate its integration into ophthalmological workflows, supporting early detection and management of retinal and systemic vascular diseases.

References

[1] Liskowski, P., & Krawiec, K. (2016). Segmenting retinal blood vessels with deep neural networks. *IEEE Transactions on Medical Imaging, 35*(11), 2369–2380. https://doi.org/10.1109/TMI.2016.2546227.

[2] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 234–241). Springer. https://doi.org/10.1007/978-3-319-24574-4$_2$8.

[3] Staal, J., et al. (2004). Ridge-based vessel segmentation in color images of the retina. *IEEE Transactions on Medical Imaging, 23*(4), 501–509. https://doi.org/10.1109/TMI.2004.825627.

[4] Hoover, A., Kouznetsova, V., & Goldbaum, M. (2000). Locating blood vessels in retinal images by piecewise threshold probing of a matched filter response. *IEEE Transactions on Medical Imaging, 19*(3), 203–210. https://doi.org/10.1109/42.845178.

[5] Jiang, Z., et al. (2018). Retinal blood vessel segmentation using fully convolutional network with transfer learning. *Computerized Medical Imaging and Graphics, 68*, 1–15. https://doi.org/10.1016/j.compmedimag.2018.04.001.

[6] Fu, H., et al. (2016). Deepvessel: Retinal vessel segmentation via deep learning and conditional random field. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 132–139). Springer. https://doi.org/10.1007/978-3-319-46723-8$_1$6.

[7] Meyer, M. C., et al. (2021). Pixel-wise deep learning for vessel segmentation in retinal images. *Scientific Reports, 11*, 1–12. https://doi.org/10.1038/s41598-021-91442-7.

[8] Li, Q., Feng, B., Xie, L., Liang, P., Zhang, H., & Wang, T. (2016). A cross-modality learning approach for vessel segmentation in retinal images. *IEEE Transactions on Medical Imaging, 35*(1), 109–118. https://doi.org/10.1109/TMI.2015.2458702.

[9] Zhang, J., et al. (2020). Attention guided U-Net for accurate retinal vessels segmentation. *Medical Image Analysis, 63*, 101789. https://doi.org/10.1016/j.media.2020.101789.

[10] Oliveira, A., Pereira, S., & Silva, C. A. (2018). Retinal vessel segmentation based on fully convolutional neural networks. *Expert Systems with Applications, 112*, 229–242. https://doi.org/10.1016/j.eswa.2018.06.030.